

2. Executive dashboard design for a given business analytics scenario using Tableau Public.

OUTCOME

After completing this experiment, the student will be able to:

- Connect datasets to Tableau Public
- Create basic KPIs
- Build visual charts
- Combine visuals into a dashboard
- Publish a dashboard online
- Understand the fundamentals of Business Intelligence dashboards

DATASET REQUIRED

Sample Superstore dataset

Download from Kaggle:

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Extract the file and use:

✓ Sample - Superstore.xls (preferred)

or

✓ superstore.csv

TASK OVERVIEW

Students will perform the following:

1. Download dataset
2. Connect dataset in Tableau Public
3. Create 3 KPI sheets
4. Create 3 chart sheets
5. Build a dashboard
6. Add filters & interactivity
7. Publish to Tableau Public

STEP-BY-STEP PROCEDURE

STEP 1: Open Tableau & Connect Data

1. Open **Tableau Public**
2. Under *Connect*, click:

- **Microsoft Excel** (for .xls)
 - **OR Text File** (for .csv)
3. Browse to the downloaded file → **Open**
 4. Drag **Orders** sheet to canvas (if not auto-loaded)
 5. Verify data types:
 - Order Date → **Date**
 - Sales → **Number (decimal)**
 - Profit → **Number (decimal)**
 - Quantity → **Number**

STEP 2: Create KPI Sheets

KPI 1 — Total Sales

1. Go to **Sheet 1** → rename to **Total Sales**
2. Drag **Sales** → **Text** on Marks Card
3. Format as Currency (0 decimals)
4. Make text large (36px, Bold)

KPI 2 — Total Profit

1. Duplicate previous sheet → rename **Total Profit**
2. Replace Sales with **Profit**
3. Format as Currency

KPI 3 — Profit Margin

1. Duplicate sheet → rename **Profit Margin**
2. Create Calculated Field:
3. $SUM([Profit]) / SUM([Sales])$
4. Drag Profit Margin % → Text
5. Format as **Percentage (1 decimal)**

STEP 3: Create Visual Charts

Chart 1 — Sales Over Time (Line Chart)

1. New Worksheet → rename **Sales Over Time**
2. Drag Order Date → Columns → change to **Month**
3. Drag Sales → Rows

4. Set Marks type → **Line**

Chart 2 — Sales by Region (Bar Chart)

1. New Worksheet → rename **Sales by Region**
2. Drag Region → Columns
3. Drag Sales → Rows
4. Set Marks → **Bar**
5. Drag Profit → Color

Chart 3 — Top 5 Products

1. New Worksheet → rename **Top 5 Products**
2. Drag Product Name → Rows
3. Drag Sales → Columns
4. Right-click Product Name → Filter → **Top** → **Top 5 by SUM(Sales)**

STEP 4: Build the Dashboard

1. Click **New Dashboard**
2. Set Size → **Desktop (1000×800)**
3. Drag KPI sheets → arrange in **one horizontal row**
4. Drag **Sales Over Time** below KPIs
5. Drag **Sales by Region** and **Top 5 Products** side-by-side below the line chart
6. Adjust sizes and alignment

STEP 5: Add Filters

Date Filter

1. In Sales Over Time sheet → drag Order Date → Filters → Range of Dates
2. Right-click Filter → Show Filter
3. Dashboard → Apply filter to **All Worksheets Using This Data Source**

Region Filter

1. In Sales by Region sheet → right-click Region → Show Filter
2. Dashboard → Apply to **All Worksheets Using This Data Source**

STEP 6: Add Interactivity

1. Dashboard → **Actions**

2. Add Action → **Filter**
3. Source Sheet: **Sales by Region**
4. Target Sheets: *All other sheets*
5. Run on: **Select**

Now clicking a region filters all charts.

STEP 7: Publish

1. File → **Save to Tableau Public As...**
2. Sign in to your Tableau Public account
3. Name the dashboard
4. Copy the public link
5. Add screenshots + link in lab report

Output :-

Experiment 2

Total Sales

Total Profit

Profit Margin

Product Name

Limit
Top 5 by SUM([Sales])

₹ 2,297,200.86

12.5%

Sales Over Time



Sales by Region



Top 5 Products

Product Name	Sales
Canon imageCLASS ..	150K
Fellowes PB500 Elec..	120K
Cisco TelePresence ..	100K
HON 5400 Series Ta..	80K